3 Block Diagram and Signal Flow Graph

INTRODUCTION:

- Block diagram is the pictorial representation of a control system representing functions performed by the every singale component and flow of signals.
- It helps in describing the cause and effect (input – output) relationship throughout the systems.
- It helps in simplifying the practical complex controls systems in smaller segments which are represented by a block diagram. Overall system can be represented by combination of these blocks.
- Transfer function: It is a ratio of Laplace transform of output to the Laplace transform of input when all the inital conditons are assumed to be zero.

$$H(s) = \left.\frac{Y(s)}{X(s)}\right|_{\text{Initial Condition}=0}$$

Where X(s): input signal
Y(s): output signal
And H(s): Transfer function

A graphical representation of this behaviour is the block diagram as shown in figure 3.2



Figure: 3.1 General block diagram

- A complete system can be arranged by interconnecting individual blocks as per requirements of the system.



Figure: 3.2 Illustration of a control system by the block diagram

In above block diagram $G_1(s)$ and $G_2(s)$ represent the transfer function of individual elements of a control system.
The system is feedback type or closed-loop type. As the output signal C(s) is fed back and compared with the input R(s).
The difference E(s) = R(s) – C(s) is the actuating signal or error signal.

Block Diagram of an RC circuit

An RC electrical system which consists of two components, resistance and capacitance.

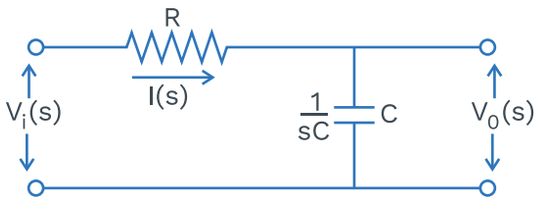


Figure: 3.3 RC circuit

If we apply the input signal $V_i(s)$ then we get the output voltage $V_0(s)$ across the capacitor. Current flowing in this circuit is given as -

$$I(s)=\frac{V_i(s)-V_0(s)}{R}$$

And

$$V_0(s)=I(s)\times\frac{1}{sC}$$

Gray Matter Alert!!!

- It is essential to show the sign of the input signal which is entering the summing point.

Block diagram representation of above RC network can be drawn as

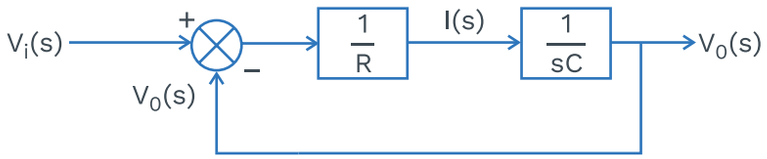


Figure: 3.4 Illustration of a RC network by the block diagram

IMPORTANT COMPONENTS:

1. Summing Point:

- It is used to sum (or difference) more than one input signals.
- It is represented by a circle with cross.
- Consecutive summing points can be interchanged, it will not alter the output signal as shown below.



Figure: 3.5 Summing Point

2. Take off Point:

- Any point is considered as take-off point if it has single input and multiple output.

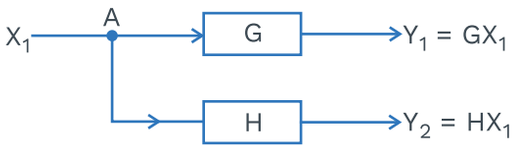


Figure: 3.6 Take Off Point

In the above figure take-off point is represented by a point 'A'.

3. Forward Gain:

- The direction of forward gain is from input towards output.

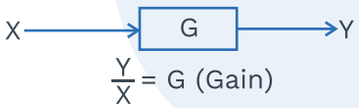


Figure: 3.7 Forward Gain

BLOCK DIAGRAM REDUCTION TECHNIQUE:

To determining the transfer function of a system, a procedure called block diagram reduction technique is followed. Some of the important rules for block diagram reduction are given below.

1. Gains Connected in cascade:

- If the output of one block is the input of next block, then these blocks are connected in cascade.
- When several blocks are connected in cascade the overall gain of resulting system is the multiplication of individual gain as illustrated below.

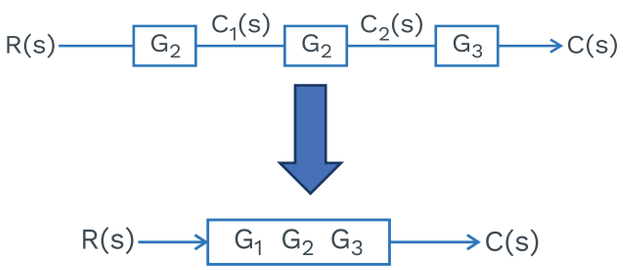


Figure: 3.8

Gray Matter Alert!!!

- In control system, while connecting blocks in cascade, loading effect is ignored i.e., forward block doesn't affect the output of preceding block.

2. Blocks Connected in Parallel:

When one or more blocks connected in parallel then we can replace these blocks by single block whose gain will be sum of these individual blocks.

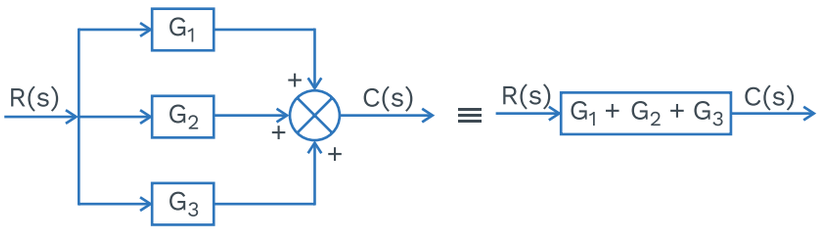


Figure: 3.9 Equivalence of Blocks in Parallel.

3. The summing point is moved after the block:

Summing point is shifted from a position before the block to a position after the block as shown below.

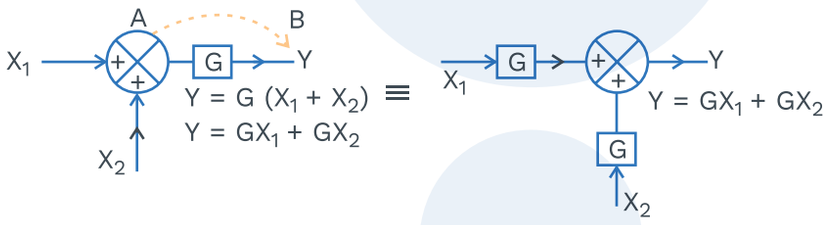


Figure: 3.10 Summing point moved after the block

4. The summing point is moved before the block:

Shifting of a summing point from a position after a block to a position before the block is shown below.

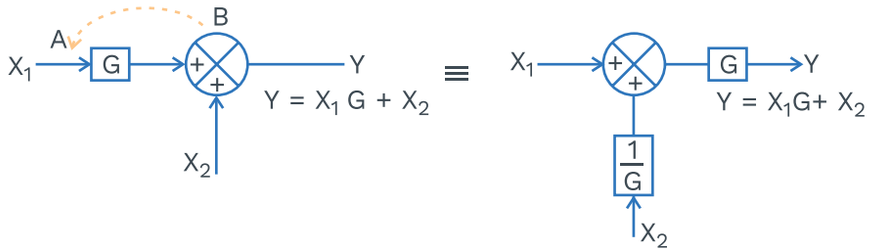


Figure: 3.11 Summing point at position B shifted to position A

5. The take off point is moved after the block:

Shifting of a take off point from a position before a block to a position after the block is shown in figure.

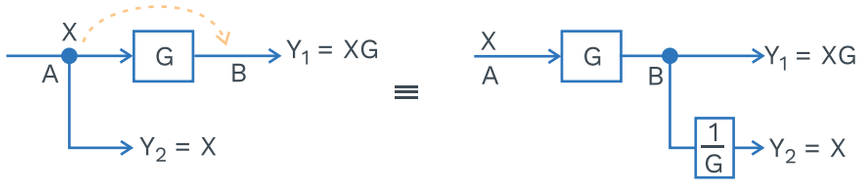


Figure: 3.12 Take off point at position A is shifted to position B

6. **The take off point is moved before the block:**
   Shifting of a take off point from a position after a block to a position before the block is shown in figure.

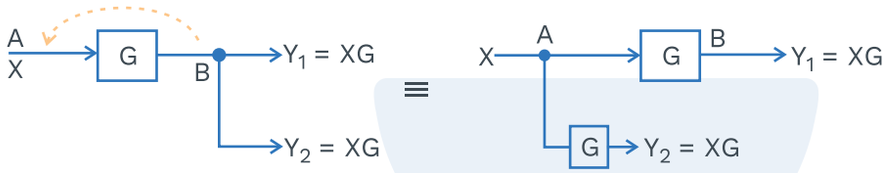


Figure: 3.13 Take off point at position B shifted to position A

7. Shifting of a take off point from a position before a summing point to a position after the summing point shown in figure.

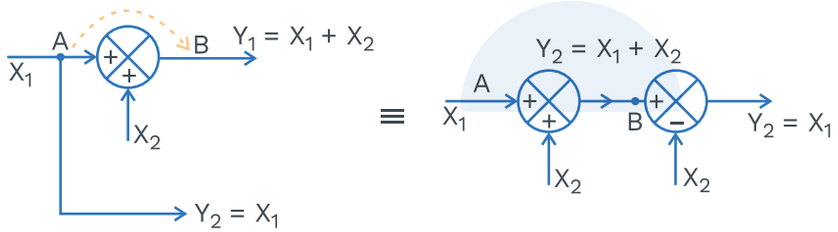


Figure: 3.14 Take off point position A is shifted to position B.

Note:

- In above case second input to new summer will have opposite sign.

8. Shifting of a take off point from a position after a summing point to a position before summing point is shown in figure.

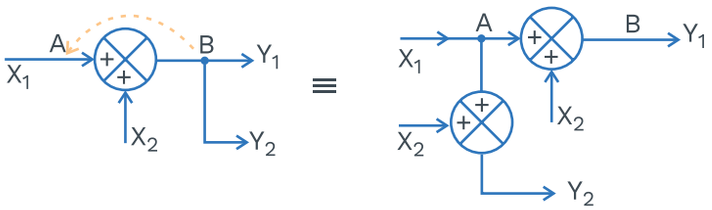


Figure: 3.15 Take off point at position B shifted to position A.

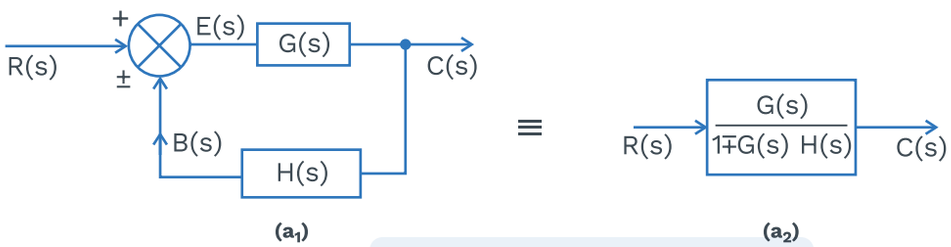
Note:

- In above case second input of new summer will have same sign.

9. Elimination of a feedback element:

(a) A closed loop control system is represented by a block diagram shown in figure 3.16 where feedback path B(s) = C(s). H(s) is compared with the input R(s) which results in an error E(s) given by

E(s) = [R(s) ± B(s)]



(a_1) (a_2)

Figure: 3.16 Elimination of feedback element in a closed loop system

From figure (a_1) $\frac{C(s)}{E(s)} = G(s)$ ..... (I)

G(s) is termed as forward path transfer function and $\frac{B(s)}{C(s)} = H(s)$ ..... (II)

H(s) is feedback path transfer function.

The output of the summing point is given by –

$$E(s) = [R(s) \pm B(s)] \quad \text{..... (III)}$$

Substituting for E(s) and B(s) in equation (III) from equation (I) and (II) the following equation is obtained

$$\frac{C(s)}{G(s)} = R(s) \pm C(s)H(s) \quad \text{..... (IV)}$$

Simplifying equation (IV) the overall transfer function denoted as T(s) is given by

$$T(s) = \frac{C(s)}{R(s)} = \frac{G(s)}{1 \mp G(s)H(s)} \quad \text{..... (V)}$$

The equivalent transfer function as obtained by equation (V) is represented by a single block diagram shown in figure (a_2).

Rack your Brain

Explain the reasons to frame the rules of block diagram reduction techniques?

(b) A unity feedback control system is shown in figure (b_1), where feedback path transfer function is unity and the equivalent of unity feedback control system to a single block representation is shown in figure (b_2)

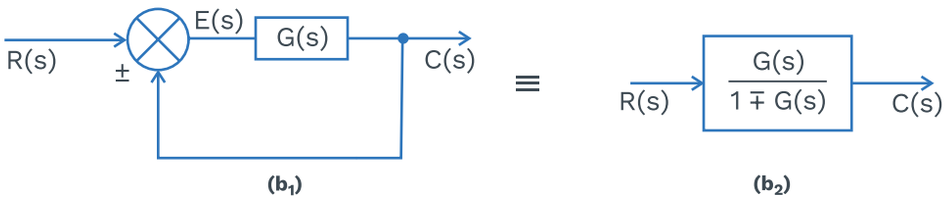


Figure: 3.17 Elimination of feedback in a unity feedback control system

since H(s) = 1, the transfer function of a unity feedback control system relating the output and input is given by-

$$\frac{C(s)}{R(s)} = \frac{G(s)}{1 \mp G(s).1}$$

$$\frac{C(s)}{R(s)} = \frac{G(s)}{1 \mp G(s)}$$

(c) The transfer function relating E(s) and R(s) for a closed loop control system shown in figure (3.18) is determined below.

As E(s) = [R(s) ± B(s)] ..... (I)

And B(s) = C(s) H(s) ..... (II)

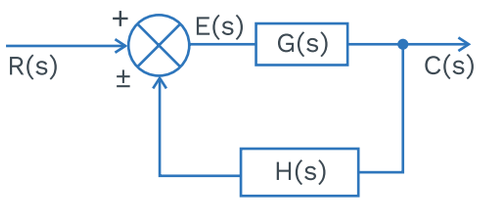


Figure: 3.18 Block diagram of a closed loop control system

Therefore, substituting the value of B(s) in equation (I) the following relation is obtained:

E(s) = R(s) ± C(s) H(s) ..... (III)

It is also seen that

C(s) = E(s). G(s)

Substitute the value of C(s) in equation (III)

$$E(s) = [R(s) \pm E(s)\, G(s)\, H(s)] \quad \text{..... (IV)}$$

Simplifying equation (IV) the transfer function relating E(s) and R(s) is obtained below :

$$\frac{E(s)}{R(s)} = \frac{1}{1 \mp G(s)H(s)}$$

Steps to solve block diagram are-

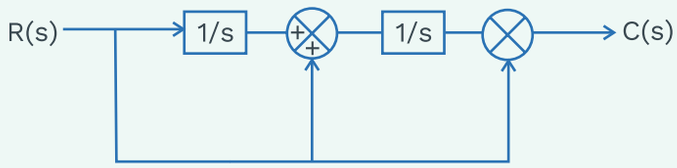
1. Combine all the cascade blocks.
2. Combine all the parallel blocks.
3. Eliminating all the minor feedback loops.
4. Shifting the summing point to the left and take off point to the right of the major loop.
5. Repeat steps 1-4 until canonical form of transfer function is achieved.

Note:-

In the objective questions it is always desirable to redraw the block diagrams so that each signal path is separated. This will simplify the block diagram.

Previous Year's Questions

For the block diagram shown in figure, the transfer function $\frac{C(s)}{R(s)}$ is equal to



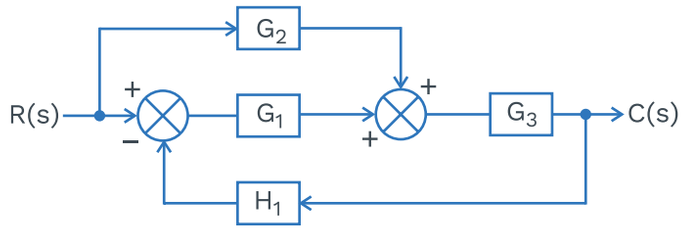
(a) $\frac{s^2+1}{s^2}$ (b) $\frac{s^2+s+1}{s^2}$

(c) $\frac{s^2+s+1}{s}$ (d) $\frac{1}{s^2+s+1}$

Ans: (b)

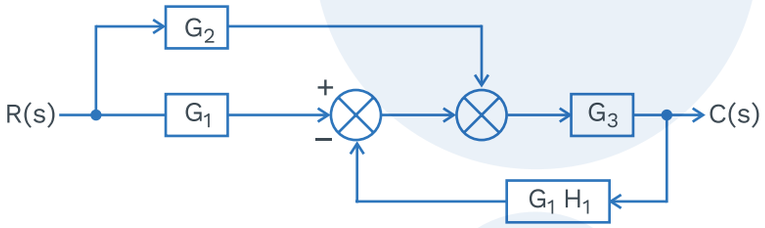
Example-1

The overall gain $\frac{C(s)}{R(s)}$ of the block diagram shown below is

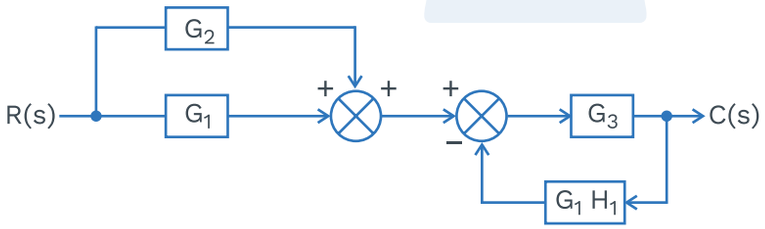


Solution:

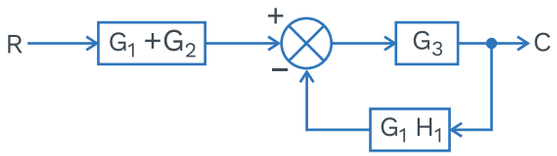
On shifting the summing point before the block G_1 to a position after block G_1



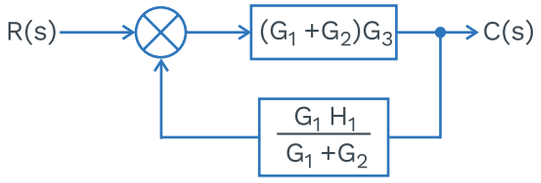
Interchange the consecutive summing point after block G_1.



Eliminate summing point after block G_1.



Shift the summing point after block $(G_1 + G_2)$ to a position before block $(G_1 + G_2)$ and combine blocks $(G_1 + G_2)$ and G_3.

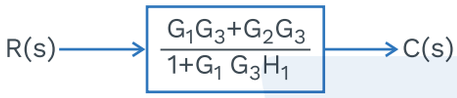


The overall gain is determined below –

$$\frac{C(s)}{R(s)} = \frac{(G_1 + G_2)G_3}{1 + (G_1 + G_2)G_3 . \frac{G_1 H_1}{(G_1 + G_2)}}$$

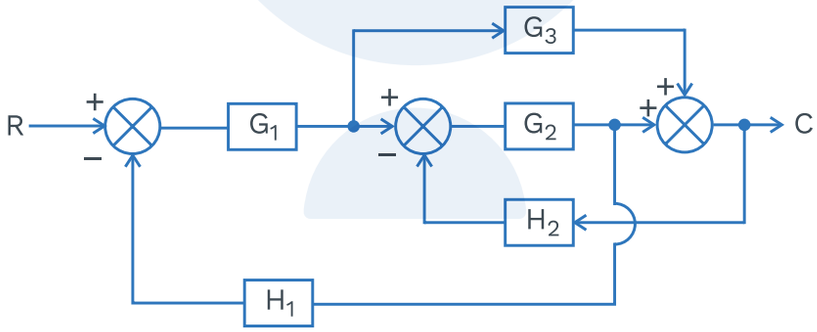
$$\frac{C(s)}{R(s)} = \frac{G_1 G_3 + G_2 G_3}{1 + G_1 G_3 H_1}$$

The single block diagram of above equation can be shown as –



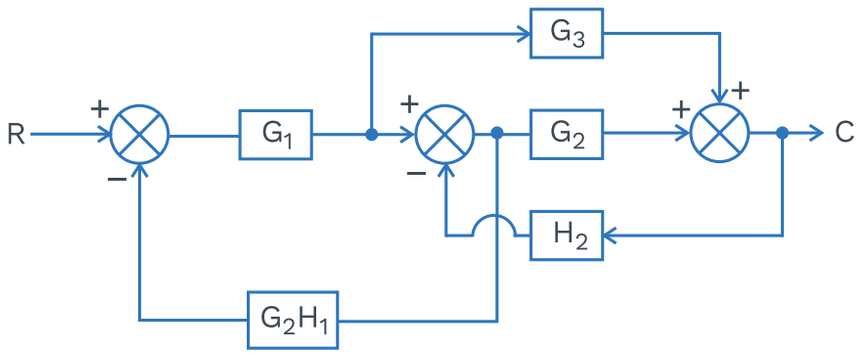
Example-2

For the given block diagram, obtain the transfer function.

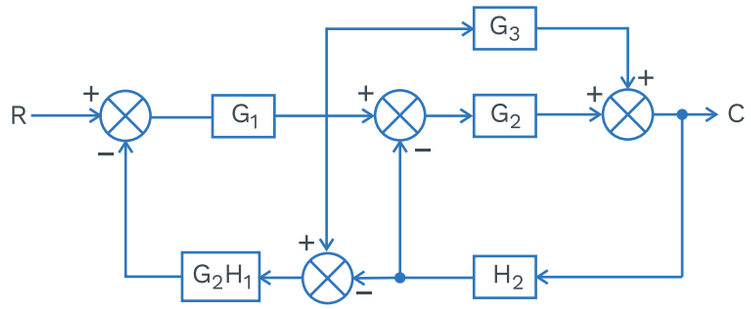


Solution:

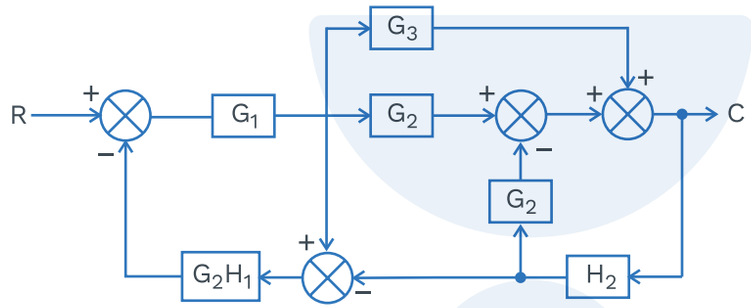
On shifting the take off point after block G_2 to a position before block G_1.



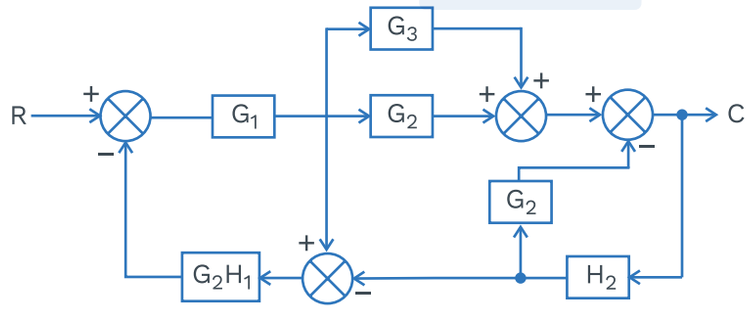
Shift the take off point before block G_2 to a position after block G_1



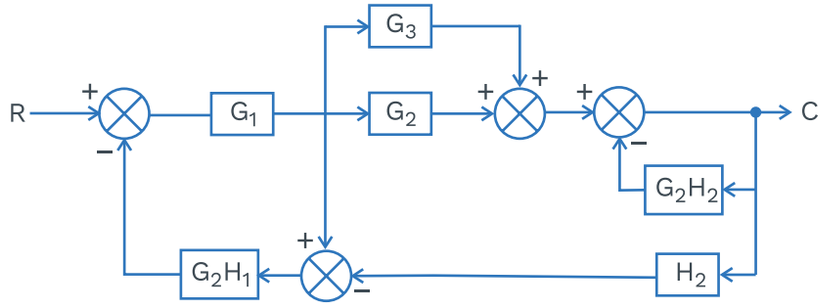
Shift summing point before block G_2 to a position after block G_2.



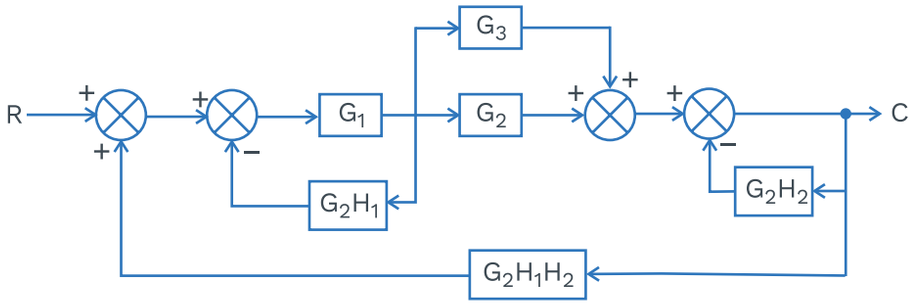
Interchange consecutive summing points after block G_2.



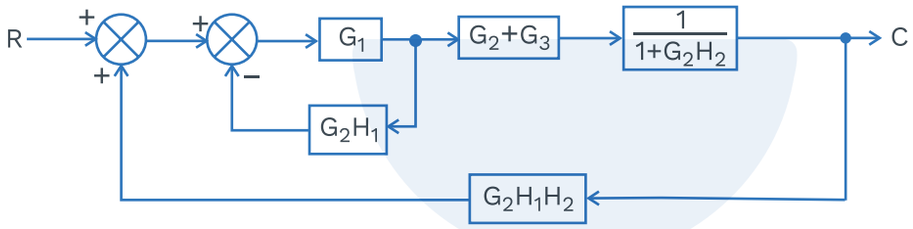
Shift the take off point after block H_2 to a position before block H_2.



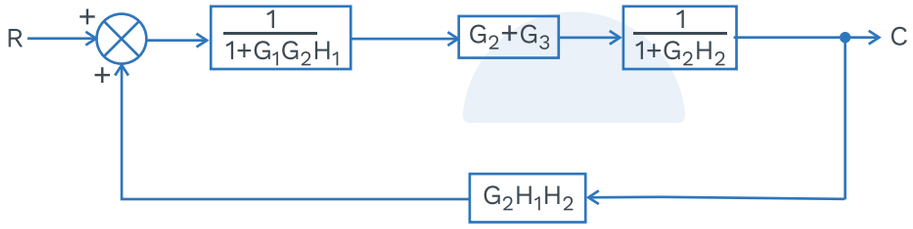
The block diagram which is shown above may be redrawn as:



Eliminate two consecutive summing points after block G_2.



Eliminate the summing point before block G_1.



The overall transfer function is determined below:

$$\frac{C}{R} = \frac{\frac{G_1}{1+G_1G_2H_1}.(G_2+G_3).\frac{1}{(1+G_2H_2)}}{1-\frac{G_1}{(1+G_1G_2H_1)}.(G_2+G_3).\frac{1}{(1+G_2H_2)}.(G_2H_1H_2)}$$

$$= \frac{G_1G_2 + G_1G_3}{1+G_2H_2+G_1G_2H_1-G_1G_2G_3H_1H_2}$$

MULTIPLE INPUT MULTIPLE OUTPUT SYSTEMS:

When there are multiple inputs in a linear system, every individual input may be treated independently of the others, the total output of the given system may be obtained by the superposition i.e. outputs that a corresponding to each input are added together.

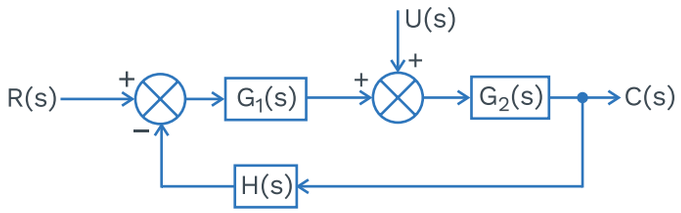


Figure: 3.19 Diagram of two input system

Consider a linear two input system shown in figure 3.19. The response to the reference input can be obtained by assuming U(s) = 0. The corresponding block diagram shown in figure 3.20.

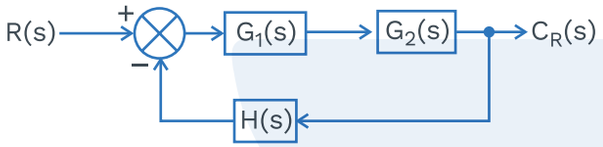


Figure: 3.20 Block diagram when R(s) acting alone

$C_R(s)$ = output due to R(s) acting alone

$$C_R(s) = \frac{G_1(s)G_2(s)}{1 + G_1(s)G_2(s)H(s)} R(s) \qquad \text{..... (I)}$$

Similarly, the response to the input U(s) is obtained by assuming R(s) = 0.

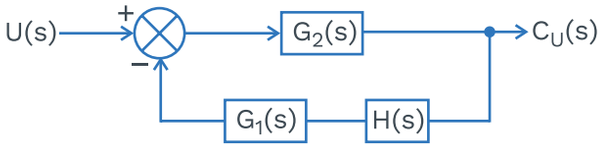


Figure: 3.21 Block diagram when U(s) acting alone

$$C_U(s) = \frac{G_2(s)}{1 + G_1(s)G_2(s)H(s)} U(s) \qquad \text{..... (II)}$$

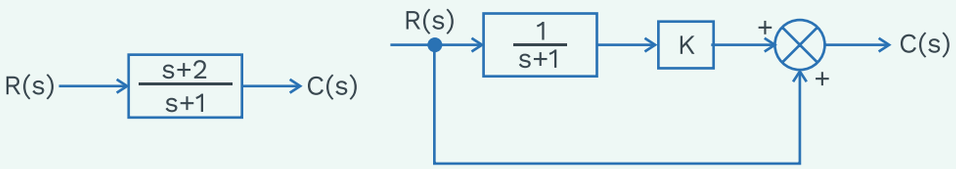
Adding both the responses, we get

$$C(s) = C_R(s) + C_U(s)$$

$$= \frac{G_2(s)}{1 + G_1(s)G_2(s)H(s)} \left[G_1(s)R(s) + U(s)\right]$$

Previous Year's Questions

For what value of K are the two block diagrams as shown below equivalent?

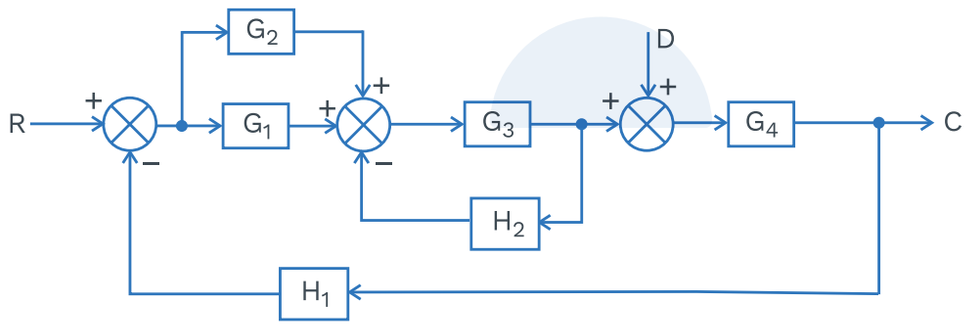


(a) 1 (b) s + 1 (c) 2 (d) s + 2

Ans: (a)

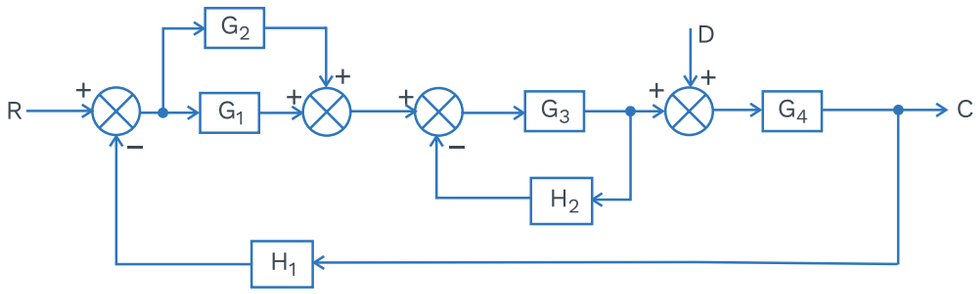
Example–3

Consider the block diagram shown below, obtain the ratio C/R, C/D and also find the total output for the given system.

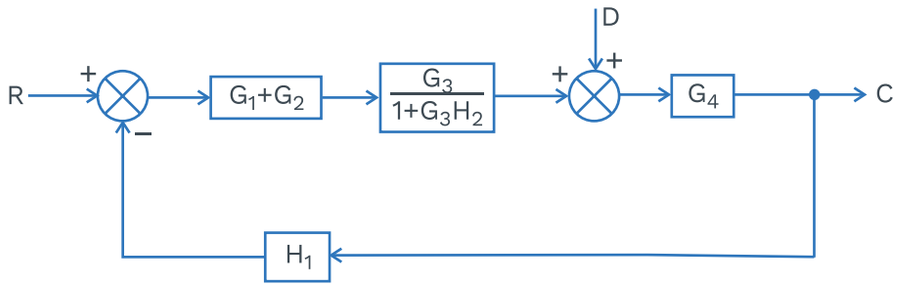


Solution:

The given block diagram can be redrawn as

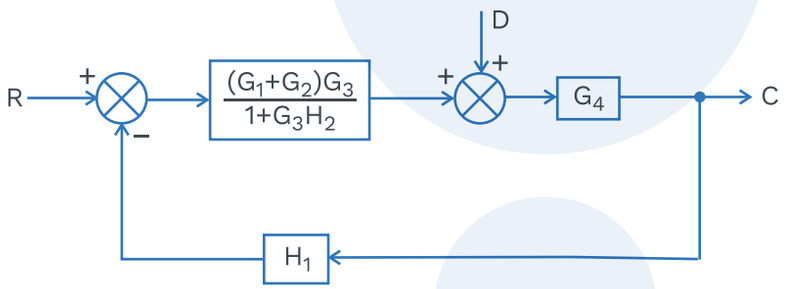


Eliminate summing points after block G_1 and before block G_3 we get

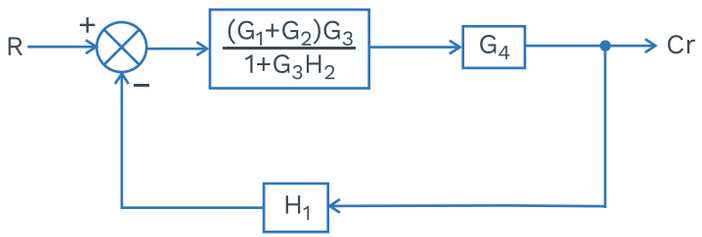


Combine blocks $(G_1 + G_2)$ and $\frac{G_3}{1+G_3H_2}$

(I) In order to determine the ratio C/R consider D = 0, corresponding block diagram is shown below -



Consider D = 0, the corresponding output is denoted as C_r.

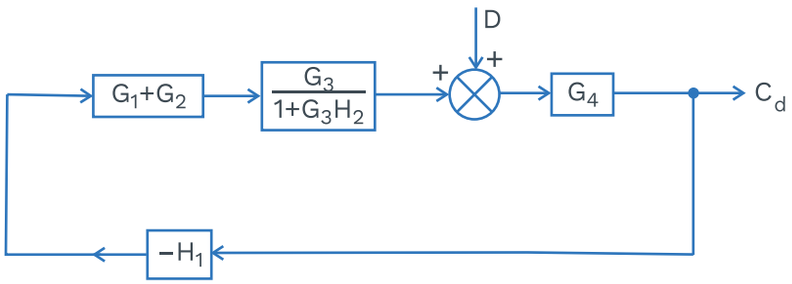


The ratio C_r/R is determined below

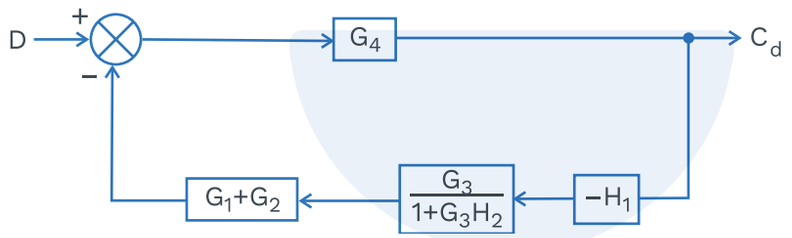
$$\frac{C_r}{R} = \frac{\frac{(G_1+G_2)G_3}{1+G_3H_2}.G_4}{1+\frac{(G_1+G_2)G_3}{(1+G_3H_2)}.G_4H_1}$$

$$= \frac{G_1G_3G_4 + G_2G_3G_4}{1+G_3H_2+G_1G_3G_4H_1+G_2G_3G_4H_1}$$

(II) Consider R = 0, the corresponding output is denoted as C_d , hence the block diagram can be drawn as



The above block diagram can be redrawn as



According to block diagram the ratio C_d/D is determined below –

$$\frac{C_d}{D} = \frac{G_4}{1 + G_4 . \frac{(G_1 + G_2) G_3 H_1}{(1 + G_3 H_2)}}$$

$$= \frac{G_4 (1 + G_3 H_2)}{1 + G_3 H_2 + G_1 G_3 G_4 H_1 + G_2 G_3 G_4 H_1}$$

The total output is given by,

$$C = C_r R + C_d D$$

$$C = \frac{G_1 G_3 G_4 + G_2 G_3 G_4}{1 + G_3 H_2 + G_1 G_3 G_4 H_1 + G_2 G_3 G_4 H_1} . R + \frac{G_4 (1 + G_3 H_2)}{1 + G_3 H_2 + G_1 G_3 G_4 H_1 + G_2 G_3 G_4 H_1} . D$$

FEED FORWARD COMPENSATION:

Let us consider the example of figure 3.22. The input U(s) in this block diagram represents disturbance input in control system. The effect of such an input is to introduce error into the system which needs to be kept low i.e., within acceptable limits. This can be achieved by compensation technique. In several systems where the disturbance input can be predicted, its effect can be eliminated by a feed forward compensation technique.

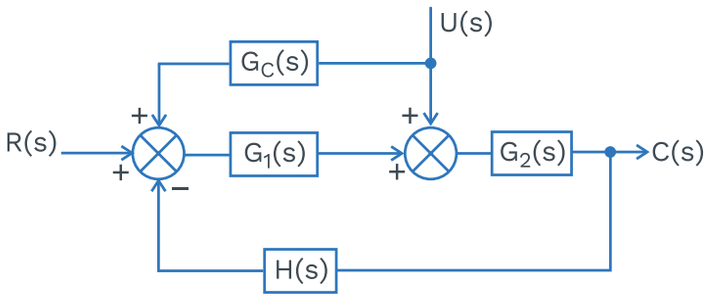


Figure: 3.22 Forward compensation

The compensating block $G_C(s)$ causes additional input of $G_C(s)$ U(s) along with U(s), then this would contribute to output a term-

$$C_U(s) = \frac{G_2(s) + G_1(s)G_2(s)G_C(s)}{1 + G_1(s)G_2(s)H(s)}U(s)$$

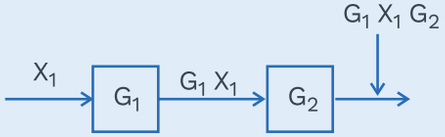
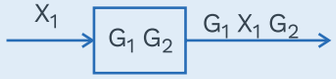
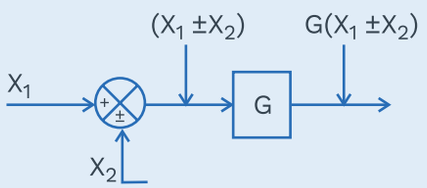
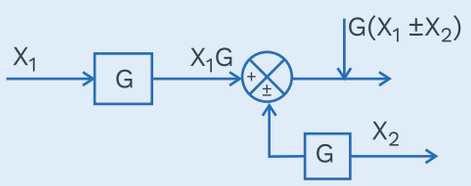
For this to cancel out the output component $C_U(s)$ due to disturbance input U(s) the following condition has to be met.

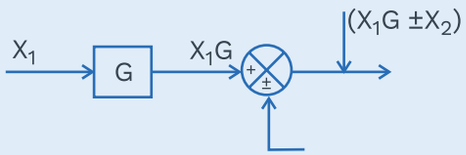
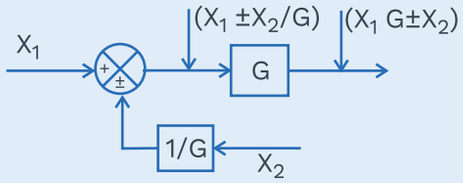
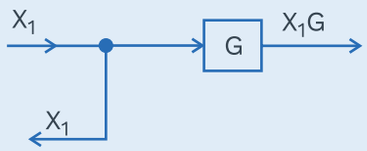
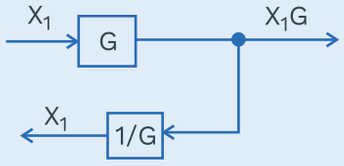
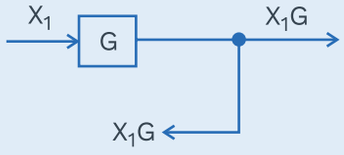
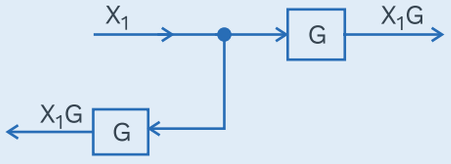
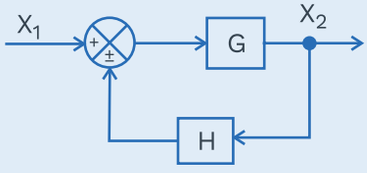
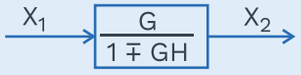
$$G_2(s) + G_1(s)\ G_2(s)\ G_C(s) = 0$$

Or $$G_C(s) = -\frac{1}{G_1(s)}$$

Rules for block diagram algebra -

The following table is illustrating the transformation theorems. The letter G, H is used to represent any transfer function and X_1, X_2 denote any transformed signals.

Rule	Original Diagram	Equivalent Diagram
1. Combining blocks in cascade		
2. Moving a summing point after a block		

Rule	Original Diagram	Equivalent Diagram
3. Moving a summing point ahead of a block		
4. Moving a take-off point after a block		
5. Moving a take-off point ahead of a block		
6. Eliminating a feedback loop		

SIGNAL FLOW GRAPH

Introduction:

- Signal flow graph displays the transmission of signal through the system. It is an easy way to draw the signals and more easier for manipulating them.
- It consists of a network with nodes representing each of the system variables linked together by the directed branches.
- A SFG may be regarded as a simplified version of block diagram and it does not require any reduction process as there is availability of a flow graph gain formula that relates the output and input system variables.

Gray Matter Alert!!!

- The signal flow graph was introduced by S.J. Mason for the cause and effect representation of linear system.

Basic Properties of signal flow graph

The important properties of signal flow graph are: -

1. It is applicable only on the linear systems.
2. Signal moves along the branches only in the direction indicated by the arrows of the branches.
3. The branch directing from node y_k to node y_j shows the dependence of y_j upon y_k but not the reverse.
4. The drawn SFG equation must be algebraic equation in the form of cause and effect.
5. Nodes are basically represented by variables. The nodes are generally arranged from the input to the output or from left to right.

Basic terminologies (elements) in signal flow graph

Some mentioned terms which are useful for the purpose of identification and execution of signal flow graph algebra are given below.

1. **Node**– Node represents the system variable, the node value is equal to the sum of all the incoming signal at that node.
   The outgoing signals from a node does not affect the values of the node variables. For example, R, E, B, F and C are nodes in figure 3.23.

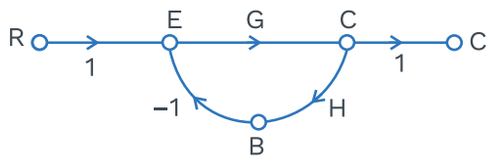


Figure: 3.23 Signal flow graph of a closed loop system

2. **Input node (source node) –**
   A node which consists of only outgoing branches is called as input node. For example, R is an input node in figure 3.23

3. **Output node or sink -**
   It is node with only incoming branches. For example the node C in figure 3.23 is an output node.

Gray Matter Alert!!!

- Block diagram gives a pictorial representation of a control system by way of short handling the transfer function.
- Signal flow graph shortens the representation of a control system by way of eliminating summing symbol, take off point and blocks.
- Apart from the differences in the physical properties of the SFG and the block diagram, the SFG is inhibited by more strict mathematical rules, while the block diagram notation is more simple to understand..

4. **Path -**
   The traversal of connected branches in the direction indicated by the branch arrows with no node traversed more than once is called as path.

Rack your Brain

What is difference between input node and initial node?

5. **Forward path –**
   It is a path between input node to output node so that no node should be repeated. For example R-E-F-C is a forward path in figure 3.23.

6. **Loop –**
   A path that emerges and closes at the same node is called as loop. For example, E-F-B-E is a loop in figure 3.23.

7. **Self loop –**
   Loop which are formed from only one branch and one node is called the self loop. For example H is self loop in figure.

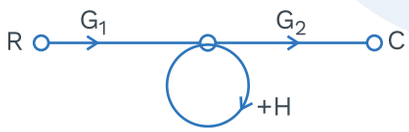


Figure: 3.24 Self loop

Gray Matter Alert!!!

- If input node is not given we cannot make input node by adding unity gain branch.
- If output node is not given, we can form output node by adding a unity gain branch.

8. **Non touching loop –**
   If a loop does not have any common node then it is referred to as non-touching loop.

9. **Forward path gain –**
   It is the product of the branch gains encountered in traversing a forward path. For example, forward path gain of the path R-E-F-C is G in figure 3.23.

10. **Loop gain –**
    It is the product of branch gains encountered in traversing the loop. For example, loop gain of the loop E-F-B-E is –GH in figure 3.23.

MASON'S GAIN FORMULA:

The relation between an output and input variable of signal flow graph is given by the net gain between the output and the input nodes which is called overall gain of the system. Mason's gain formula for determining overall system's gain is given by,

$$T = \frac{1}{\Delta}\sum_{k} P_k \Delta_k$$

Where, P_k = path gain of kth forward path

Δ = determinant of the graph

= 1 – (sum of loop gains of all individual loops) + (sum of gain products of all possible combination of two non-touching loops) – (sum of gain products of all possible combination of three non touching loops) ...,

Δ_k = the value of Δ for the part of the graph not touching the kth forward path (path factor).

T = overall gain of the system.

Rules for drawing signal flow graph –

1. The signal travels along a branch in the direction of an arrow.
2. The input signal is multiplied by the transmittance to obtain the output signal.
3. Input signal at a node is sum of all signals entering at that node.
4. A node transmits signals in all branches leaving that node.

Rack your Brain

What is the limitation of Mason's Gain formula?

Conversion of block diagrams into signal flow graph –

These steps need to be followed to convert a block diagram into signal flow graph.

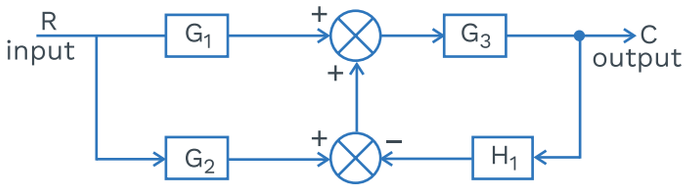
1. Represent all the variables,signals,take off points and summing points of the block diagram as nodes in SFG.
2. Represent the blocks of the block diagram as branches in SFG.
3. Represent the transfer functions which lies in the blocks of block diagram as gains of the branches in SFG.
4. Connect the node according to the block diagram. If there is connection between two nodes and there is no block between them, then magnitude of one is represented as the gain of the branch.

Gray Matter Alert!!!

Self loop at initial node (of loop gain unity or non-unity) can always be ignored. We can may or may not make initial node as input node.

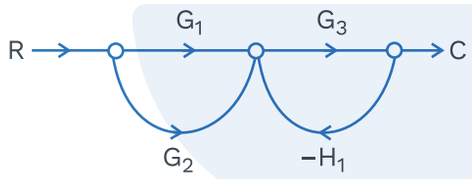
Example-4.

Draw a signal flow graph for the system whose block diagram is given below and evaluate closed loop transfer function relating the output and input.



Solution:

As per the block diagram the corresponding signal flow graph is as follows–



The two forward paths are

$$P_1 = G_1 G_3$$

$$P_2 = G_2 G_3$$

The only individual loop is

$$L_1 = - G_3 H_1$$

The closed loop touches both the forward paths, therefore path factors are

$$\Delta_1 = 1 \text{ and } \Delta_2 = 1$$

The graph determinant is

$$\Delta = 1 - (L_1) = 1 - (-G_3 H_1)$$

$$\Delta = 1 + G_3 H_1$$

Applying Mason's gain formula the closed loop transfer is

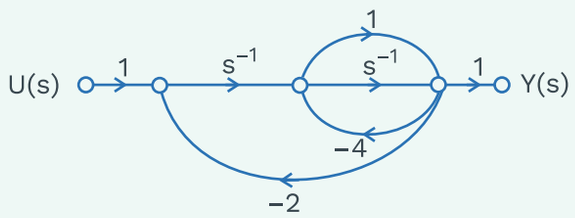
$$\frac{C}{R} = \frac{P_1 \Delta_1 + P_2 \Delta_2}{\Delta}$$

$$= \frac{G_1 G_3 .1 + G_2 G_3 .1}{1 + G_3 H_1}$$

$$= \frac{G_1 G_3 + G_2 G_3}{1 + G_3 H_1}$$

Previous Year's Questions

The signal flow graph for a system is given below. The transfer function $\frac{Y(s)}{U(s)}$ for this system is



(a) $\frac{s+1}{5s^2+6s+2}$

(b) $\frac{s+1}{s^2+6s+2}$

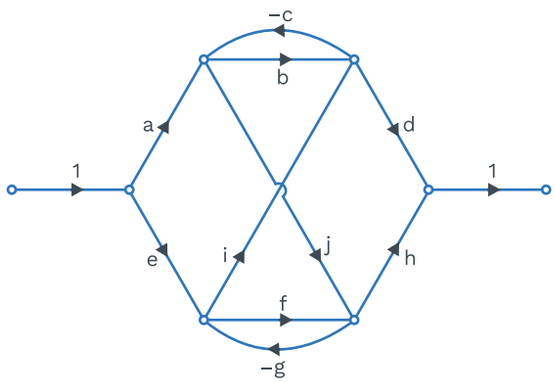
(c) $\frac{s+1}{s^2+4s+2}$

(d) $\frac{1}{5s^2+6s+2}$

Ans: (a)

Example-5

For the signal flow graph shown in figure, what will be the graph determinant (Δ)?



Solution:

$L_1 = -bc$, $L_2 = -fg$, $L_3 = jgic$, $L_1L_2 = bcfg$

$\Delta = 1 - (-bc - fg + cigi) + bcfg = 1 + bc + fg - cigj + bcfg.$

Example-6

Obtained the overall transfer function $\frac{C}{R}$ for the signal flow graph shown below.



Solution:



For the given SFG we cannot directly apply Mason gain formula because R is not input node. So, we have to create a input node by dragging unity gain to node X and the graph is like as follows –



First, we have to calculate $\frac{R}{X}$

Forward path, $p_1 = 1$

Loop, $L_1 = H_1$

$L_2 = H_2$

$$\frac{R}{X} = \frac{p_1\Delta_1}{\Delta} = \frac{1(1-H_2)}{\Delta} = \frac{(1-H_2)}{\Delta} \quad \text{..... (I)}$$

Note: Self loop at input with any gain can be eliminated for the calculation of transfer function.

For calculating $\frac{C}{X}$

$p_1 = G$

$L_1 = H_1$

$L_2 = H_2$

$$\frac{C}{X} = \frac{p_1\Delta_1}{\Delta} = \frac{G(1-0)}{\Delta}$$

$$\frac{C}{X} = \frac{G}{\Delta} \qquad \text{..... (II)}$$

By dividing equation (II) and (I) we get –

$$\frac{C}{R} = \frac{G}{1 - H_2}$$

Example-7

For the given SFG find the value of X_2/X_1?



Solution:

We cannot apply Mason gain formula because X_1 is not input node.

We know that,

Node value is equal to the sum of incoming signal.

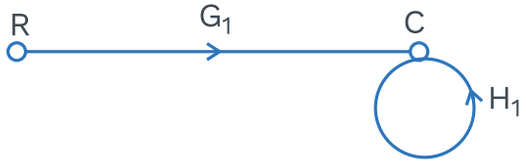
So at node X_2,

$$X_2 = 4X_1$$

$$\frac{X_2}{X_1} = 4$$

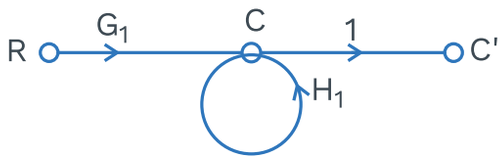
Example-8

The signal flow graph of a system is shown below where C' is the output and R is the input find transfer function $\frac{C}{R}$.



Solution:

We can form a output node C' by adding unity gain branch.



Forward path, $P_1 = G_1$

Loop $L_1 = H_1$

$\Delta_1 = 1$

$\Delta = 1 - H_1$

$$TF = \frac{C'}{R} = \frac{P_1 \Delta_1}{\Delta} = \frac{G_1.1}{1 - H_1}$$

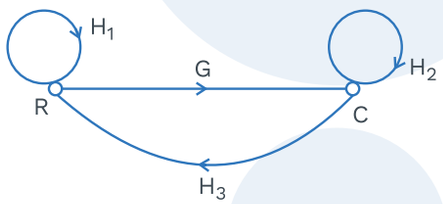
$$\frac{C'}{R} = \frac{G_1}{1 - H_1}$$

And $C' = C$ {from SFG}

So, $$\frac{C}{R} = \frac{G_1}{1 - H_1}$$

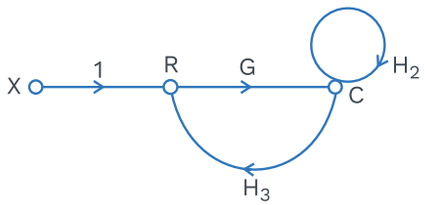
Example-9

For the given SFG, find the value of $\frac{C}{R}$



Solution:

First we have to make a input node.



$$\frac{R}{X} = \frac{P_1 \Delta_1}{\Delta} = \frac{1(1 - H_2)}{\Delta} \quad \text{..... (I)}$$

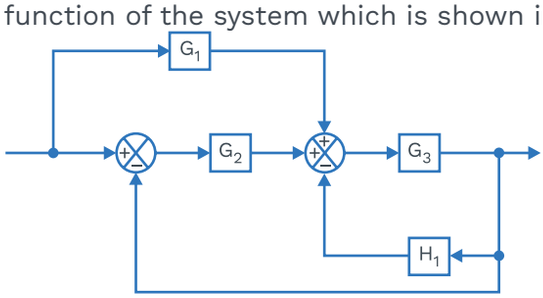
$$\frac{C}{X} == \frac{G(1 - 0)}{\Delta} = \frac{G}{\Delta} \quad \text{..... (II)}$$

By diving both equation

$$\frac{C}{R} = \frac{G}{1 - H_2}$$

Example-10

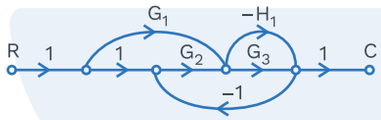
What will be the transfer function of the system which is shown in figure below?



Solution:

$P_1 = G_2G_3$, $P_2 = G_1G_3$

$L_1 = -G_3H_1$, $L_2 = -G_2G_3$, $\Delta_1 = \Delta_2 = 1$.



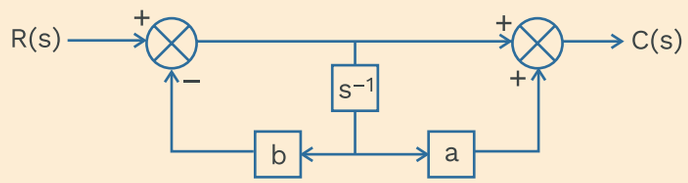
$$T(s) = \frac{G_2G_3 + G_1G_3}{1 + G_3H_1 + G_2G_3}$$

Note:-

If a SFG has the self loop at the starting node then we can remove this self loop it does not effect the gain of SFG.

PRACTICE QUESTIONS

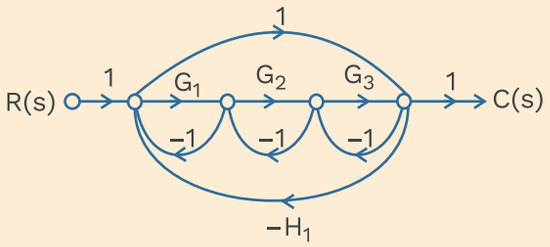
Q1 **The figure shows the block diagram of a control system. Calculate the transfer function C(s)/R(s)?**



(a) $\frac{s+a}{s-b}$ **(b)** $\frac{s-a}{s+b}$ **(c)** $\frac{s+a}{s+b}$ **(d)** $\frac{s-a}{s-b}$

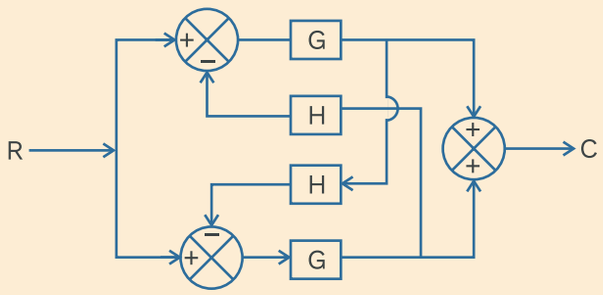
Sol. **Correct answer is (b).**

Q2 **The signal flow graph is shown in figure. The transfer function C(s)/R(s) is-**



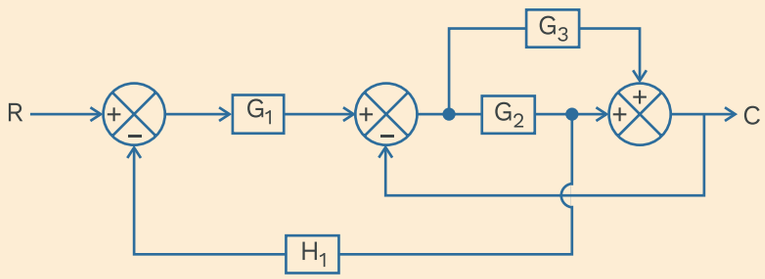
Sol. $\frac{G_1G_2G_3 + 1(1+G_2)}{2 + G_1 + G_3 + H_1 + G_2 + G_1G_2G_3H_1 + G_1G_3 + G_2H_1}$

Q3 **The overall transfer function of the system shown in figure is-**



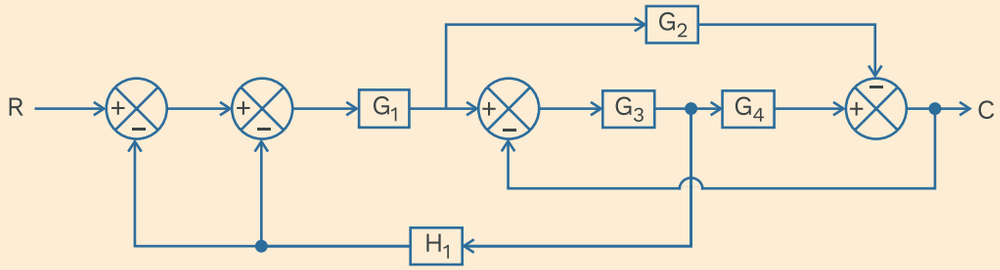
Sol. $\frac{C}{R} = \frac{2G}{1 - GH}$

Q4 **Find $\frac{C}{R}$ from the block diagram shown.**



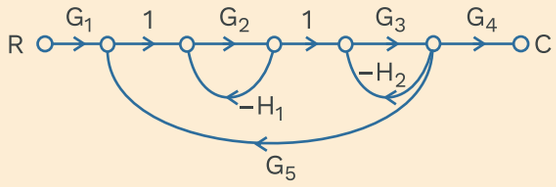
Sol. $\frac{C}{R} = \frac{G_1(G_2 + G_3)}{1 + G_2 + G_3 + G_1G_2H_1}$

Q5 **Draw the signal flow graph from the given block diagram and determine the overall transfer function.**



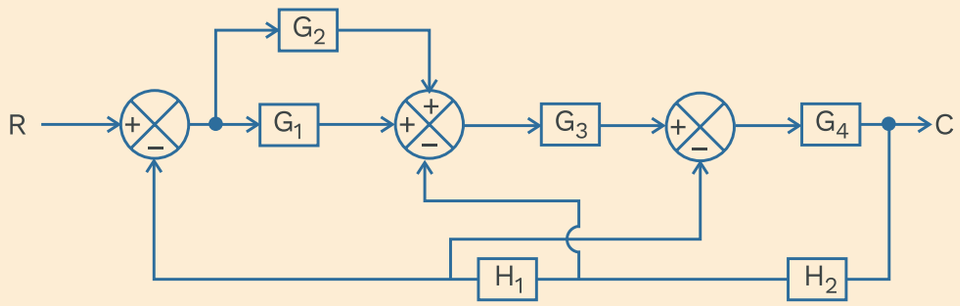
Sol. $$\frac{C}{R} = \frac{G_1G_3G_4 - G_1G_2}{1 + G_3G_4 + G_1G_3H_1 + G_1G_2 + G_3H_1}$$

Q6 **The signal flow graph represents a feedback control system. Determine the overall transfer function using reduction technique of block diagram.**



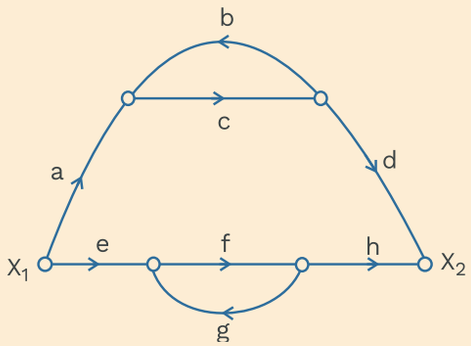
Sol. $$\frac{C}{R} = \frac{G_1G_2G_3G_4 + G_1G_4G_5(1 + G_2H_1)}{1 + G_2H_1 + G_3H_2 + G_2G_3H_1H_2}$$

Q7 **Obtain the overall transfer function for a system represented by the block diagram.**



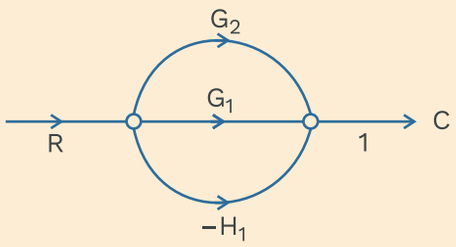
Sol. $$\frac{G_1G_3G_4 + G_2G_3G_4}{1 + G_3G_4H_2 + G_4H_1H_2 + G_1G_3G_4H_1H_2 + G_2G_3G_4H_1H_2}$$

Q8 **The signal flow graph shown below has two parallel paths and two isolated loops. Find the overall transmittance which relates X_2 and X_1.**



Sol. $$\frac{efh(1 - bc) + acd(1 - gf)}{1 - (bc + gf) + (bcgf)}$$

Q9 **For the given signal flow graph determine the overall transmittance relating C and R.**



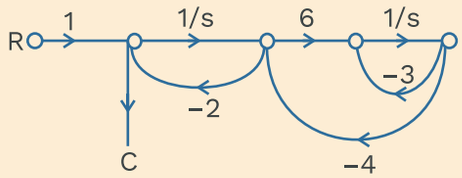
Sol. $\dfrac{G_1 + G_2}{1 + G_1H_1 + G_2H_1}$

Q10 **Represent the given set of equations by a SFG and find the overall gain relating X_5 and X_1.**

$$X_2 = aX_1 + fX_2 \quad X_3 = bX_2 + eX_4$$
$$X_4 = cX_3 + hX_5 \quad X_4 = dX_4 + gX_2$$

Sol. $\dfrac{X_5}{X_1} = \dfrac{abcd(1) + ag(1 - ce)}{1 - f - ce - dh + fec + fdh}$

Q11 **The signal flow graph of a second order system is given below –**



Calculate the transfer function of the given system

Sol. $\dfrac{C}{R} = \dfrac{s(s + 27)}{s^2 + 29s + 6}$